

Strengths:

Modular Approach: The app uses HTML for structure, CSS for design, and JavaScript for logic, following standard web architecture. This modular approach makes it easier to maintain, update, or expand the app in the future.

User-Friendly Interface: The design is visually appealing and intuitive, with a clean layout, clear input fields, and readable sections for weather details. Responsive elements and attractive styling (e.g., background, rounded containers, large fonts) enhance the user experience across devices.

Dynamic Interactivity: JavaScript enables real-time updates of weather information based on user input without reloading the page. This improves responsiveness and creates a smoother, more modern feel.

Clear Data Presentation: Weather metrics (temperature, description, humidity, pressure, wind speed/direction) are displayed in distinct, easy-to-read boxes, ensuring users can quickly understand the information provided.

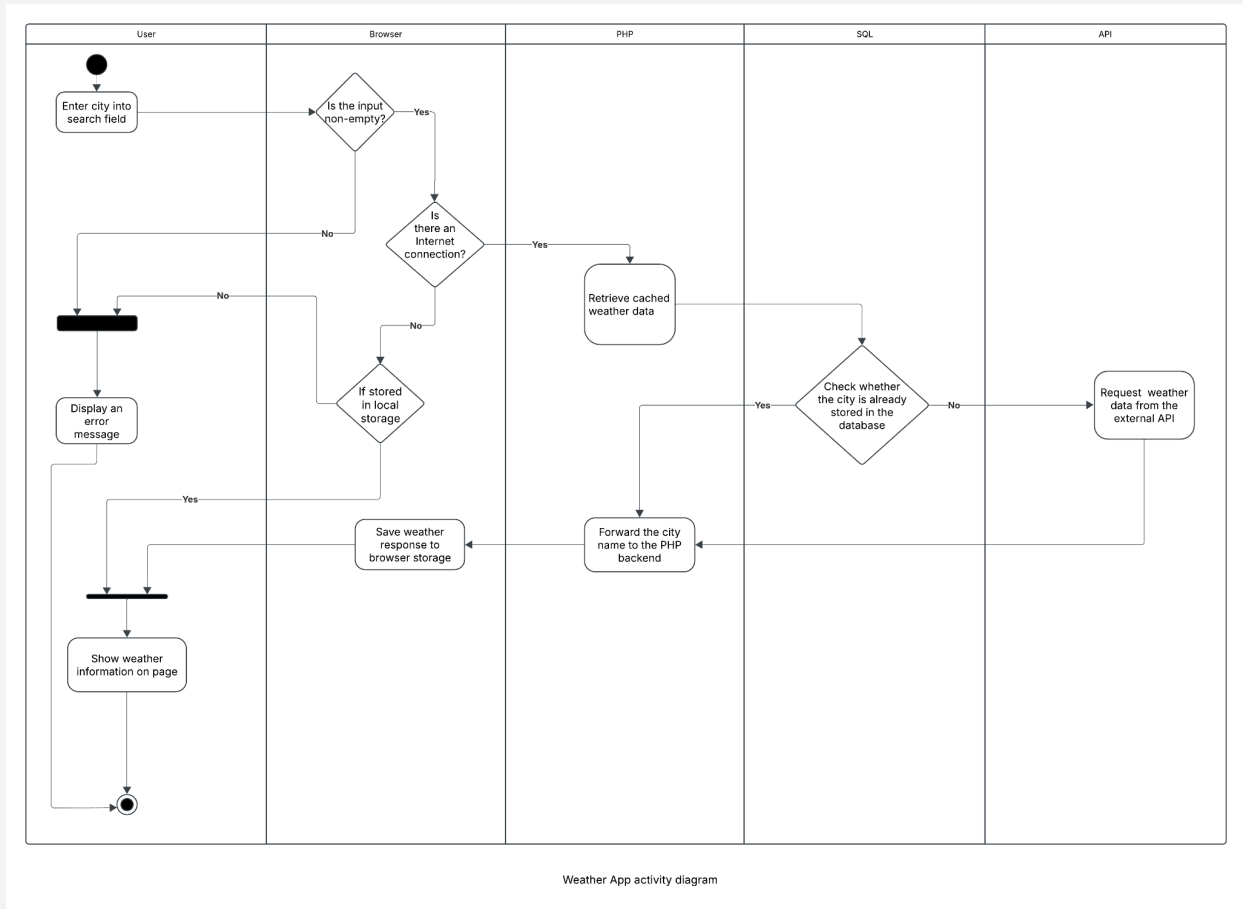
Broad Compatibility: By relying on standard web technologies, the app is accessible on most browsers and devices.

Weaknesses:

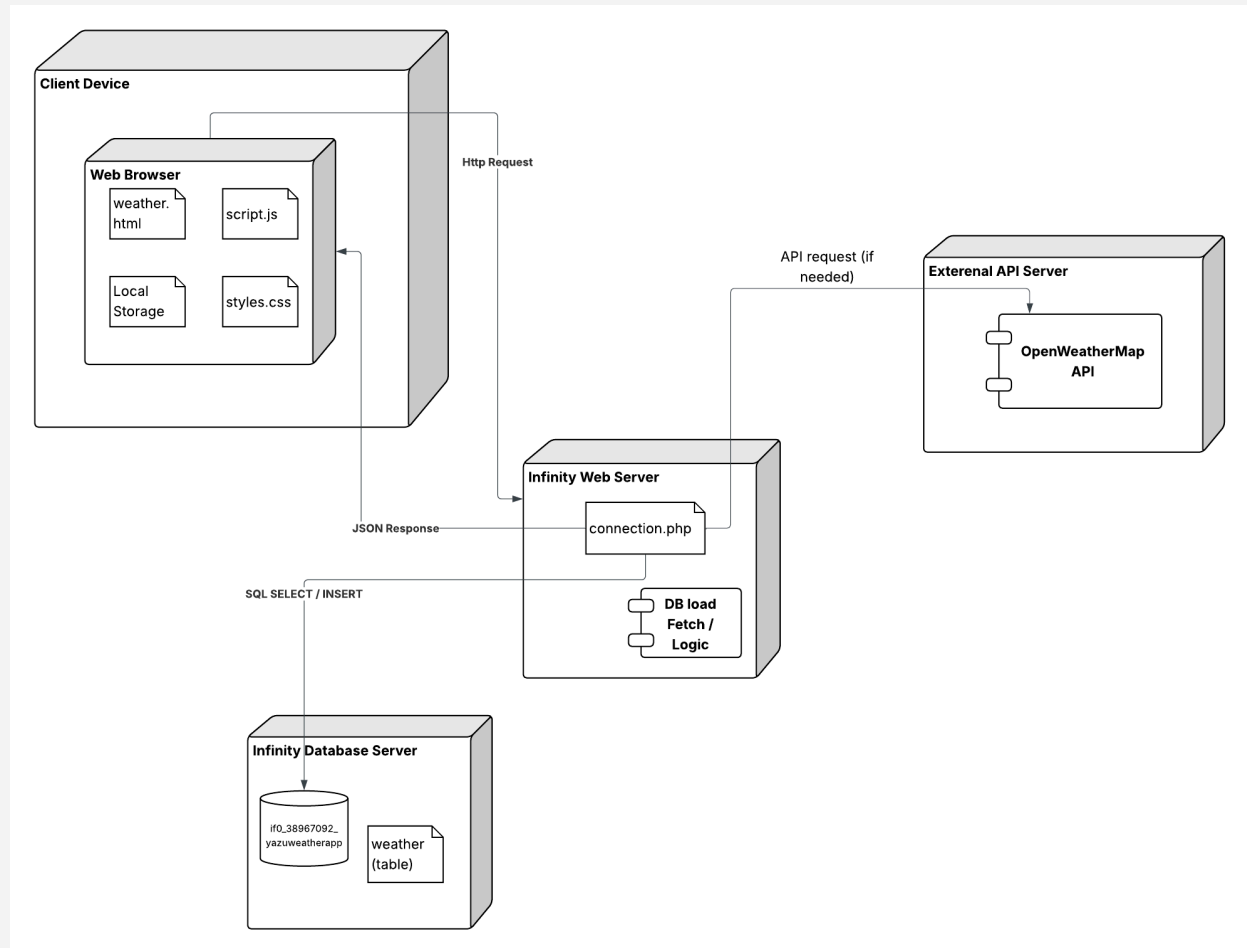
- **Limited Error Handling:** The frontend currently lacks robust mechanisms to inform users of errors such as invalid city names, network issues, or empty results, which could lead to confusion or frustration.
- **Security Concerns:** Direct communication between the frontend and backend (PHP) may expose sensitive endpoints or API keys if not properly protected. There is also a risk of backend vulnerabilities if input is not validated or sanitized.
- **No Data Caching:** The architecture appears to fetch fresh data for every request without caching, which may result in redundant API calls, slower performance, and unnecessary load on the backend or external APIs.
- **Lack of Advanced Features:** The app does not currently support features like search history, favorites, or location auto-detection, which could make it more engaging and practical for users.

UML Diagrams :

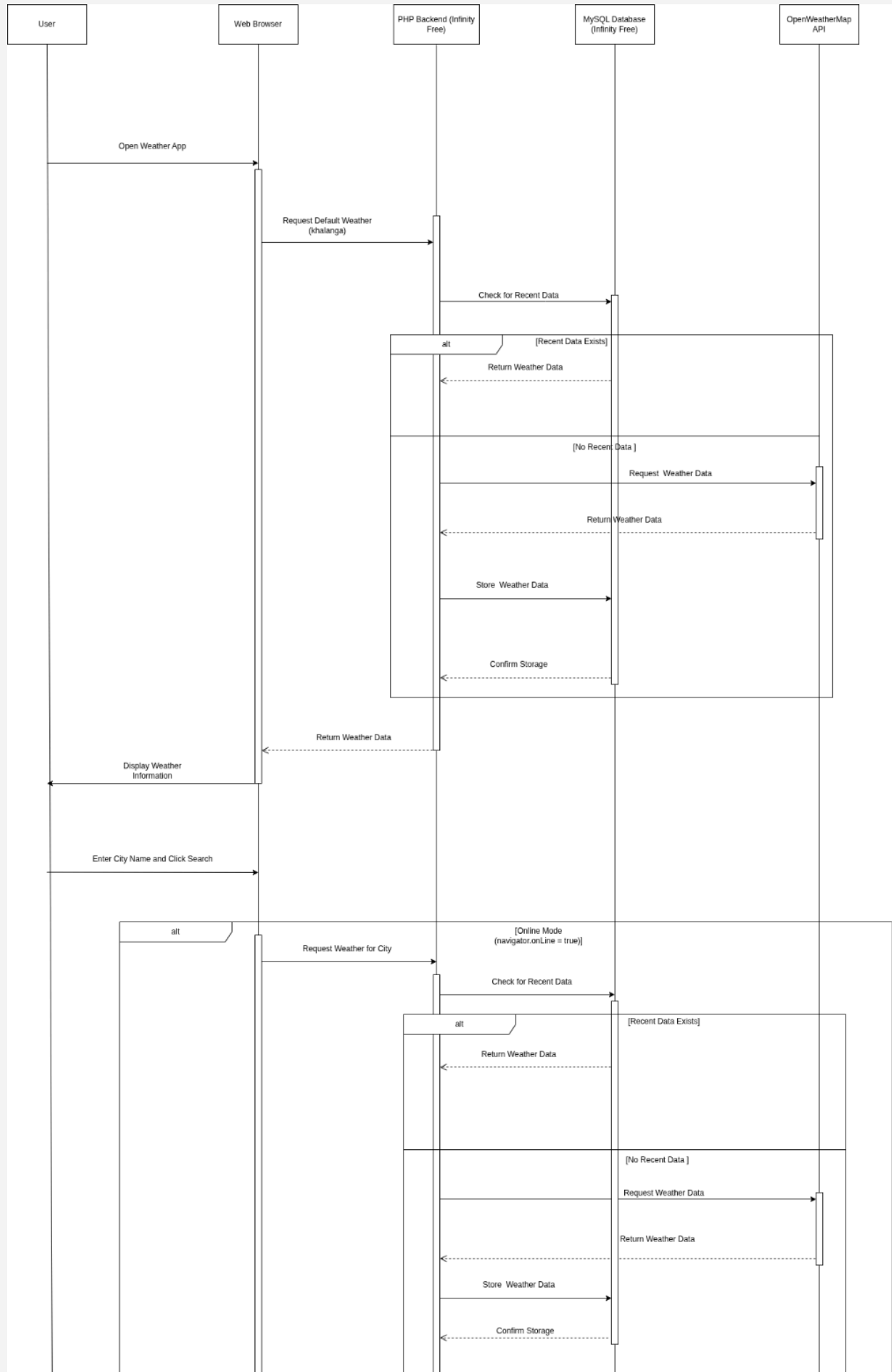
Activity Diagram



Deployment Diagram:



Sequence Diagram:



Hosted Link

Youtube Video Link